Misspecification: DR vs structural NLL

Final deck — setup first, define before use

OPC project

2026

A controlled simulator study: when the true click model leaves a bilinear class, does propensity-aware policy optimization (DR) pull ahead of matched logged likelihood (NLL), relative to a no-misspec baseline?

How to read this deck

1. **Part A — Objects.** Catalog, embeddings, clicks, logger, value.
2. **Part B — Experiment setup.** Build a one-parameter family of worlds (intensity dial). Longest part: every symbol defined before use.
3. **Part C — Train & measure.** Restricted score classes, two oracles, DR vs NLL, difference-in-differences.
4. **Part D — Locked run.** Gates and confirmatory numbers.

Catalog

Finite recommendation catalog:

- users $\mathcal{U}$, with count $n_U = |\mathcal{U}|$;
- items $\mathcal{A}$, with count $n_A = |\mathcal{A}|$;
- embedding dimension d (locked d=32).

A **policy** π assigns, to each user u, a distribution $\pi(\cdot \mid u)$ over items. The success metric is click-through rate (CTR) under that policy.

Two embedding tables

From a BPR factorization of organic data we get **clean** matrices

$$X^{\mathrm{c}} \in \mathbb{R}^{n_U \times d}, \qquad Y^{\mathrm{c}} \in \mathbb{R}^{n_A \times d},$$

with rows $\boldsymbol{x}_u^{\mathrm{c}}$ (user) and $\boldsymbol{y}_a^{\mathrm{c}}$ (item).

Optional **representation bias** then produces the matrices that the logger and all learners actually see:

$$X \in \mathbb{R}^{n_U \times d}, \qquad Y \in \mathbb{R}^{n_A \times d},$$

with rows $\boldsymbol{x}_u$ and $\boldsymbol{y}_a$.

Rule: truth uses clean vectors; learning uses biased vectors. That mismatch is held fixed across the intensity dial below—it is *not* the residual we introduce.

Click probabilities (baseline world)

True click probability

$$q^\star(u,a) \;=\; \sigma\big(\ell^\star(u,a)\big), \qquad \sigma(z) = \frac{1}{1+e^{-z}}.$$

In the **baseline** world (no residual yet), the true logit is bilinear in the clean vectors:

$$\ell_0^\star(u,a) \;=\; \alpha\, \tilde{z}_1(u,a) \;+\; b_0.$$

Here $\tilde{z}_1$ is the clean score $z_1(u,a) = \langle \boldsymbol{x}_u^{\mathrm{c}}, \boldsymbol{y}_a^{\mathrm{c}} \rangle$ standardized over catalog pairs, $\alpha > 0$ sets the slope, and b_0 sets the level. (How α and b_0 are chosen: Part B.)

Logging policy and logged data

User traffic weights: a prior μ on $\mathcal{U}$ (normalized). The logger scores with **biased** vectors and temperature $T_{\log} > 0$:

$$\pi_0(a \mid u) \;=\; \frac{\exp\!\big(\langle \boldsymbol{x}_u, \boldsymbol{y}_a \rangle / T_{\log}\big)}{\sum_{a' \in \mathcal{A}} \exp\!\big(\langle \boldsymbol{x}_u, \boldsymbol{y}_{a'} \rangle / T_{\log}\big)}.$$

One logged row is (u, a, r, p) with $u \sim \mu$, $a \sim \pi_0(\cdot \mid u)$, $r \sim \mathrm{Bernoulli}(q^\star(u, a))$, and $p = \pi_0(a \mid u)$ stored exactly.
Learners never see $q^\star$ or clean vectors—only (u, a, r, p) and (X, Y).

Policy value and scores

Value of a policy:

$$V(\pi) \;=\; \sum_{u} \mu(u) \sum_{a} \pi(a \mid u)\, q^{\star}(u, a).$$

A **score** is any map $f : \mathcal{U} \times \mathcal{A} \to \mathbb{R}$. Its greedy policy picks $\arg\max_a f(u, a)$; we write $V_{\mathrm{gr}}(f)$ for that CTR.

Primary study numbers are exact V_{gr} on the simulator (not doubly-robust estimates of value).

What changes with the intensity dial?

We want a one-parameter family of worlds, later written with dial $\gamma \geq 0$, such that:

1. observed features (X, Y) and the logger recipe stay fixed;
2. only the true click field $q^\star$ changes with γ;
3. at γ=0, truth is bilinear in *clean* features;
4. as γ grows, truth becomes hard for likelihood fits that are bilinear (or affine-bilinear) in the *biased* features learners see;
5. a feature-free noise control of comparable strength must *not* open the same value-vs-likelihood gap (else we added only variance).

Who sees what

Actor	Sees
Truth / residual construction	clean X^{c}, Y^{c}; residual field; $q^{\star}$
Logger	biased X, Y; temperature $T_{\log}$
Learners	biased X, Y; logged (u, a, r, p) only
Class oracles (Phase 0a)	biased X, Y; exact $q^{\star}$ (no sampling noise)

Locked feature cell "combined-high": all three bias types (warp, group, vector) at level `high`, calibrated so biased vectors keep roughly half the clean ranking signal.

Representation bias $\neq$ residual

- **Bias** changes which features the logger and learners see. It is present at every intensity, including the baseline.
- **Residual** changes the true conditional click probabilities while those observed features stay fixed.

Because bias already makes learners' features imperfect at baseline, absolute DR$-$NLL at high intensity is not the right estimand. We will use a **difference-in-differences**: how much that gap *widens* when we turn the residual on (Part C).

Preview: two oracles and a gap

Inside a restricted score class $\mathcal{F}$ on biased features, define (formally in Part C):

- **value oracle:** f maximizing $V(\pi_f)$;
- **likelihood oracle:** f maximizing expected logged Bernoulli likelihood of $q^\star$ under π_0.

Their greedy-CTR gap

$$\Delta_{\mathrm{proj}} \;=\; V_{\mathrm{gr}}(f^{\mathrm{val}}) \;-\; V_{\mathrm{gr}}(f^{\mathrm{lik}})$$

is the population signal that misspecification opens a wedge between "fit clicks under the logger" and "rank for value."

Setup goal: make Δ_{proj} grow with the intensity dial, with noise controls staying near flat.

Baseline calibration (γ=0)

Per seed, calibrate the clean bilinear world:

1. Choose slope α so that average best-item CTR $\mathbb{E}_u[\max_a q^\star(u,a)]$ equals a target (locked 0.30).
2. Equivalently write $\ell_0^\star = \alpha\,\tilde{z}_1 + b_0$ with $\tilde{z}_1$ unit-sd over catalog pairs (so $s_{\text{env}} = \alpha/\mathrm{sd}(z_1)$ in code).
3. Choose offset b_0 so mean $q^\star$ on a fixed **reference sample** of pairs equals 0.05.

The reference sample is *not* the study logger's traffic; it is a frozen calibration set used only to pin the CTR level.

Design requirements for the residual field ϕ

We will add a field $\phi(u, a)$ and dial it with intensity γ: intended logit form

$$\ell^{\star}_{\gamma}(u, a) \;=\; \alpha(\tilde{z}_1(u, a) + \gamma\, \phi(u, a)) + b'(\gamma).$$

Requirements on ϕ (construction map on the next slides):

1. **Feature-dependent** — a policy that sees features can exploit it.
2. **Hard for linear-in-features scores** — low correlation with bilinear / affine-bilinear fits on the clean (and, ideally, biased) features.
3. **Not absorbable by intercepts** — remove user/item means.
4. **Unit catalog-pair scale** — so γ is a relative sd dial (residual logit sd $\approx \gamma\alpha$ when $\mathrm{sd}(\phi) = 1$).

Why not feature-free noise?

Pure noise (even low-rank noise independent of embeddings) can hurt likelihood fits, but no feature-based policy class can systematically exploit it—so DR should not open a recoverable value gap.

That idea becomes the **NC1** control later: factored Gaussian noise, double-centered, intended same scale, no dependence on $\boldsymbol{x}^{\mathrm{c}}, \boldsymbol{y}^{\mathrm{c}}$.

Construction map

Requirement	Construction step
Feature-dependent	random projections of clean embeddings
Off BPR axes; rank-m	matrices $G, H \in \mathbb{R}^{d \times m}$
Hard for linear scores	even coordinate transform ψ
Not absorbable by intercepts	double centering
Calibrated strength	normalize; dial with γ

Step 1 — random projections G and H

Fix rank m (locked m=4) and a seed-derived RNG. Draw

$$G, H \in \mathbb{R}^{d \times m}, \qquad G_{jk}, H_{jk} \overset{\text{iid}}{\sim} \mathcal{N}(0, 1).$$

Why these matrices (not "orthogonal to z_1")

1. **Rotate off BPR axes.** Applying ψ in the raw BPR basis would make the residual depend on an arbitrary coordinate system.
2. **Separate user and item directions.** User-side G and item-side H are independent, so the interaction is not a rewrite of z_1.
3. **Rank m.** Controls complexity; factored form stays cheap.

Approximate decorrelation with z_1 is measured, not enforced (the code's "project orthogonal" hook is currently a no-op).

Step 2 — standardized projected coordinates

$$\widetilde{T}_u = X^{\mathrm{c}}\, G, \qquad \widetilde{T}_a = Y^{\mathrm{c}}\, H,$$

then per coordinate $k = 1, \ldots, m$,

$$t_{u,k} = \frac{\widetilde{T}_{u,k}}{\mathrm{sd}_{u'}(\widetilde{T}_{u',k})}, \qquad t_{a,k} = \frac{\widetilde{T}_{a,k}}{\mathrm{sd}_{a'}(\widetilde{T}_{a',k})}.$$

Why standardize? Later ψ has a kink at $|t|{=}1$. Unit-sd coordinates put that kink at one standard deviation for every coordinate, so no single projection dominates by scale alone.

Interlude — the bilinear placebo (where PL stops)

If we formed

$$z_2(u,a) \;=\; m^{-1/2}\sum_{k=1}^{m} t_{u,k}\, t_{a,k} \;=\; (\boldsymbol{x}_u^{\mathrm{c}})^{\top}(m^{-1/2}\, G\, D_u D_a H^{\top})\boldsymbol{y}_a^{\mathrm{c}},$$

this would be an *exact* rank-m bilinear form on clean features (free M on clean features can represent it exactly).

Family **PL** stops here (skip ψ): same construction style and intended magnitude, still bilinear—checks that nonlinearity, not just "extra bilinear mass," drives the oracle gap. (PL uses its own RNG stream keyed by family name; not the identical draw.)

Step 3 — even transform ψ (defined here)

Locked choice (sharpness fixed at 1):

$$\psi(t) \;=\; \tanh(t^2 - 1).$$

Properties used:

- **Even:** $\psi(-t) = \psi(t)$.
- **Nonlinear:** not a linear rescaling of t.

Why even helps. Projected coordinates are roughly symmetric under $t \mapsto -t$. Then $\psi(t)$ is even while any linear feature of t is odd, so $\mathrm{Corr}(\psi(t), t) \approx 0$. An odd map such as t^3 retains $\mathrm{Corr}(t^3, t) = 3 \neq 0$ and leaks into bilinear fits.

Worked sketch (m=1): evenness changes rankings

Suppose one coordinate with values $t_u \in \{0, 1, 2\}$ and $t_a \in \{-2, 0, 2\}$. Then $\psi(t) = \tanh(t^2 - 1)$ gives approximately

$$\psi(0) \approx -0.76, \ \psi(\pm 1) = 0, \ \psi(\pm 2) \approx 0.995.$$

Items with t_a=+2 and t_a=−2 get the *same* residual contribution; any linear score in t_a would give them opposite signs. So ϕ can flip rankings among items that look similar under z_1.

Step 4 — residual before centering

Apply ψ to each coordinate, then take matching products (this is *not* $\psi(z_2)$):

$$g_{u,k} = m^{-1/2}\psi(t_{u,k}), \qquad h_{a,k} = \psi(t_{a,k}),$$

$$\phi_{\mathrm{raw}}(u, a) \;=\; \sum_{k=1}^{m} g_{u,k}\, h_{a,k}.$$

In words: ϕ_{raw} is **bilinear in even feature maps** of the clean embeddings. An MLP on features can in principle approximate those maps (Phase 2); a plain bilinear map on raw features cannot.

Step 5 — double centering

Intercepts in rich score classes can absorb per-user or per-item means. Remove them with uniform catalog centering:

$$\phi_{\mathrm{dc}}(u, a) = \phi_{\mathrm{raw}}(u, a) - \bar{\phi}_{u\cdot} - \bar{\phi}_{\cdot a} + \bar{\phi}_{\cdot\cdot}$$

- Item means $\approx$ popularity shifts (absorbable by item intercepts).
- User means cancel in a softmax over items (ranking-invariant).

Means are unweighted over the catalog (not by μ or π_0).

Step 6 — normalize, then dial

Intended normalization: divide so catalog-pair sd equals 1,

$$\phi \;=\; \frac{\phi_{\mathrm{dc}}}{\mathrm{sd}(\phi_{\mathrm{dc}})}.$$

Then the complete true logit is

$$\ell^{\star}_{\gamma} \;=\; \alpha(\tilde{z}_1 + \gamma\,\phi) + b'(\gamma).$$

- α: per-seed slope from best-item CTR calibration (fixed across γ).
- γ: residual intensity (0, $\gamma^{\star}/2$, $\gamma^{\star}$).
- $b'(\gamma)$: re-solved on the same reference sample so mean $q^{\star}$ stays 0.05 (best-item CTR is allowed to drift).

Logger freeze and common random numbers (CRN)

Freeze $T_{\log}$. At γ=0, sharpen temperature so the logger earns a locked share of its own greedy CTR (locked share 0.8). Reuse that same $T_{\log}$ at every γ (actions stay comparable).

CRN (common random numbers). Across γ, reuse the same users, the same logged actions, and the same uniforms $U_i \sim \mathrm{Unif}(0,1)$. Clicks are

$$r_{i,\gamma} \;=\; \mathbf{1}\big[U_i < q^{\star}_{\gamma}(u_i, a_i)\big].$$

Only probabilities—hence only some realized clicks—change with γ. Purpose: variance reduction for paired contrasts, not identification by itself.

One-change controls

$R^\star$ (primary) even maps on second-index projections, then DC + scale.

PL skip ψ; keep bilinear z_2-style field at intended same scale. Isolates nonlinearity.

NC1 feature-free factored Gaussian field, DC + intended same scale. Should not open a recoverable value–likelihood gap.

R-square (design foil; not in Phase 0a run) ψ on standardized *main* score $\tilde{z}_1$, then DC.

Phase 0a measures $\Delta_{\mathrm{proj}}(\gamma) - \Delta_{\mathrm{proj}}(0)$ and aborts if $R^\star$ fails pre-registered gates.

Setup checklist

1. Clean BPR; optional combined-high bias $\rightarrow$ logger/learner features.
2. Calibrate α, b_0 at γ=0; freeze $T_{\log}$.
3. Draw G, H; form $t_{u,k}, t_{a,k}$; apply ψ; DC; normalize $\rightarrow \phi$.
4. For each γ: build $\ell^{\star}_{\gamma}$, resolve $b'(\gamma)$, draw clicks under CRN.
5. Train only on biased features + logged rows.

Locked cell (after setup is defined)

Choice	Value
Dataset	ML-1M
Feature cell	combined-high
Residual family	$R^\star$ (m=4, $\psi(t) = \tanh(t^2 - 1)$)
Intensities	$\gamma \in \{0, 0.25, 0.5\}$, $\gamma^\star$=0.5
Sample size	n=10^5 logged rows; 20 HPO trials
Confirmatory seeds	300–309

Restricted score classes on biased features

Oracles and learners share frozen $(\boldsymbol{x}_u, \boldsymbol{y}_a)$. Three classes (not a nested chain—blob intercepts need not lie in affine-bilinear):

$$\begin{aligned}
\mathcal{F}_{\text{bilin}} &= \{f = \boldsymbol{x}_u^\top M \boldsymbol{y}_a + c\}, \\
\mathcal{F}_{\text{blob}} &= \{f = \boldsymbol{x}_u^\top M \boldsymbol{y}_a + \kappa_a + c\}, \\
\mathcal{F}_{\text{aff}} &= \{f = \boldsymbol{x}_u^\top M \boldsymbol{y}_a + \boldsymbol{w}^\top \boldsymbol{y}_a + \boldsymbol{v}^\top \boldsymbol{x}_u + c\}.
\end{aligned}$$

OPC linear maps land in $\mathcal{F}_{\text{aff}}$. Published BLOB's point estimate lands in $\mathcal{F}_{\text{blob}}$.

Softmax policy from a score

$$\pi_f(a \mid u) \;=\; \frac{\exp\big(f(u,a)\big)}{\sum_{a'} \exp\big(f(u,a')\big)}.$$

Terms constant in a cancel. Positive rescaling of f sharpens π_f without changing $\arg\max_a f$.

Value oracle and likelihood oracle

Value (exact; no logged data):

$$J_{\mathrm{val}}(f) = V(\pi_f), \qquad f_\gamma^{\mathrm{val}} \in \arg\max_{f \in \mathcal{F}} J_{\mathrm{val}}(f).$$

Likelihood (infinite logger data):

$$J_{\mathrm{lik}}(f) = \sum_u \mu(u) \sum_a \pi_0(a \mid u)\, \mathrm{CE}(q^\star(u,a), \sigma(f(u,a))),$$

with $\mathrm{CE}(q,p) = -q \log p - (1-q)\log(1-p)$. Hyperparameters for f^{lik} are chosen by J_{lik}, not by greedy value.

Projection gap

$$\Delta_{\mathrm{proj}}(\gamma) \;=\; V_{\mathrm{gr}}(f_\gamma^{\mathrm{val}}) \;-\; V_{\mathrm{gr}}(f_\gamma^{\mathrm{lik}}).$$

Large when the best likelihood fit ranks badly for value. Gate G2 asks $\Delta_{\mathrm{proj}}(\gamma^\star) - \Delta_{\mathrm{proj}}(0) \geq 2.5$ CTR pt (locked mean ≈ 6.53 under $R^\star$).

OPC linear policy class

Shared corrections on biased vectors, init at identity:

$$\boldsymbol{u}' = (I + D_u)\boldsymbol{x} + \boldsymbol{b}_u, \qquad \boldsymbol{a}' = (I + D_a)\boldsymbol{y} + \boldsymbol{b}_a,$$

$$f_\theta(u, a) = s\,\langle \boldsymbol{u}', \boldsymbol{a}' \rangle, \qquad \pi_\theta = \mathrm{softmax}(f_\theta / T_{\log}).$$

Starts at the logger's ranking features ($D, \boldsymbol{b} \leftarrow 0$).

Raw DR vs logged NLL

DR (primary). With IW $w = \pi_\theta(a \mid u)/p$ and cross-fit $\hat{q}$,

$$\widehat{V}^{\mathrm{DR}} = \sum_{a'} \pi_\theta(a' \mid u)\, \hat{q}(u, a') + w(r - \hat{q}(u, a)).$$

Maximize the mean; select by clipped DR lower bound on validation.

NLL (matched baseline).

$$\mathcal{L}_{\mathrm{NLL}} = -r \log \sigma(z_\theta) - (1 - r) \log(1 - \sigma(z_\theta))$$

on the logged action's logit z_θ. No $\hat{q}$, no propensities; select by validation NLL. Shared random-search configs pair the contrast.

DiD estimand

Per confirmatory seed and intensity γ,

$$D(\gamma) = V_{\mathrm{gr}}(\pi_\gamma^{\mathrm{OPC/DR}}) - V_{\mathrm{gr}}(\pi_\gamma^{\mathrm{NLL}}), \qquad \mathrm{DiD} = D(\gamma^\star) - D(0).$$

Go: mean DiD ≥ 0.5 CTR pt, CI lower bound > 0, $\geq 7/10$ same sign, $|\mathrm{DiD}|$ under NC1 ≤ 0.3. Why DiD? Bias alone can make $D(0) < 0$ (NLL wins at baseline). The claim is that the residual *widens* DR's relative advantage.

Phase 2 and BLOB-class (same estimand)

Phase 2: matched transform $\boldsymbol{u}' = (I + D)\boldsymbol{x} + \boldsymbol{b} + \mathrm{MLP}_{64}(\boldsymbol{x})$ for OPC (DR) vs CausE-cap (likelihood-style).
BLOB-class lean check: train $f = s(\boldsymbol{x}^\top M \boldsymbol{y} + \kappa) + b$ under DR and under NLL (not the published variational ELBO)—asks whether BLOB's high absolute level is the score class or the loss/prior.

Phase 0a — oracle gates

Gate	Criterion	At $R^\star$, $\gamma^\star$, high
G1	$\geq 20\%$ users change argmax vs $\gamma{=}0$	mean 0.91
G2	$\Delta\Delta_{\mathrm{proj}} \geq 2.5$ pt	mean 6.53
G3	value$-$logger ≥ 9 pt	22.2
G4	$\lvert\Delta V_{\log}\rvert \leq 1$ pt; ESS $\pm 20\%$	pass
G5	NC1 $\lvert\Delta\Delta_{\mathrm{proj}}\rvert \leq 0.5$	max abs 0.34

PL at the same cell widens Δ_{proj} by only ~ 2.1 pt (below G2)—nonlinearity matters beyond extra bilinear mass.

Phase 1 — linear DiD (go)

$$\widehat{\mathrm{DiD}} = +1.53 \text{ pt}, \quad \mathrm{CI}\,[+0.97, +2.09], \quad 10/10 \text{ signs}.$$

γ	OPC %	NLL %	D (pt)
0	16.99	18.06	-1.07
0.25 ($= \gamma^\star/2$)	17.26	17.96	-0.70
0.5 ($= \gamma^\star$)	18.42	17.96	$+0.46$

NLL is flat across γ; OPC rises. NC1 DiD $+0.20$. OPC$-$DM DiD $\approx +1.17$ (not "DR $\approx$ DM").

Phase 2 & BLOB-class

Matched mlp64 vs CausE-cap: $\mathrm{DiD} = +2.32$ $[+1.11, +3.54]$, $9/10$. Absolute at $\gamma^\star$: OPC 19.55%, CausE 18.86%, published BLOB 20.09%.

BLOB-class $\times\{\mathrm{DR}, \mathrm{NLL}\}$ (5 seeds): $\mathrm{DiD} = +1.31$ $[+0.12, +2.49]$, $4/5$; DR levels $16.1\% \rightarrow 17.4\%$—far below published BLOB.

Claim / non-claim

Claim

On this locked residual world, matched-capacity raw DR gains on matched logged NLL as $\gamma : 0 \to \gamma^\star$, with NC1 near-null and go checks passed. Setup opens Δ_{proj} via even feature maps on a second projection index; DiD tracks that mechanism.

Do not claim

DR fixes confounding; generalization beyond ML+$R^\star$; OPC-mlp vs CausE-linear as objective evidence; beating published BLOB in absolute level; realism of $R^\star$; perfect variance matching of controls in the shipped factor-scaling code (gates use measured Phase 0a gaps).